

Numerical Data Model (Medical Cost Prediction)

a. General Information on Dataset

Dataset Name	Medical cost personal dataset
Task	Predicting charges
Target Variable	charges
Sample Count	1065
Duplicate Samples	1 duplicate row
Final Sample Count	1064
Missing Values Handled	18 missing values in the bmi column
Training/Testing Split	Train: 80%, Test: 20%
Validation Split	cross-validation used during hyperparameter (GridSearchCV)

b. Implementation Details

Feature Names (Numerical)	age, children, bmi
Feature Names (Categorical)	gender, smoker, region
Categorical Encoding	One-Hot Encoding
Final Feature Dimension	11 features (3 numerical + 8 one-hot encoded)
Feature Scaling	Numerical features scaled using StandardScaler

Cross-Validation & Hyperparameters In KNN

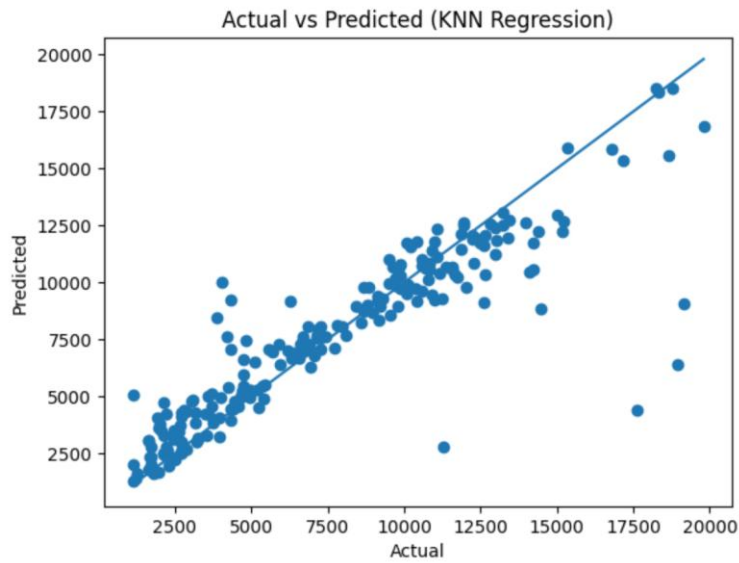
Cross-Validation	Yes, 5-fold CV (cv=5) with GridSearchCV
Hyperparameter	n_neighbors (range 1 to 29), weights='distance'

c. Results Details

Algorithm	KNN Regressor	Linear Regression
Mean Squared Error (MSE)	4277064.95	3591045.75
Root Mean Squared Error (RMSE)	2068.11	1895.01

Graphical Results

- KNN



- Linear Regression

